Practical 4 - Understanding Data with graphs

Problem statement - Used the same dataset used in Practical 2:

Question 1. Identify all duplicate cases using prof. Using all observations, find the average and standard deviation for age. Repeat the analysis by first filtering the data set to include one observation for each instructor with a total number of observations restricted to 94.

Question 2. Using a bar chart, demonstrate if instructors teaching lower-division courses receive higher average teaching evaluations.

Question 3. Plot the relationship between age and teaching evaluation scores.

Question 4. Using gender-differentiated scatter plots, plot the relationship between age and teaching evaluation scores.

Question 5. Create a scatterplot of age and evaluation scores, differentiated by gender and tenure.

Solution - Here is the complete code.

Conclusion - The TeachingRatings dataset was analyzed to identify duplicate instructors, calculate age statistics, and examine the relationship between instructor characteristics and teaching evaluations. The average age was around **48 years**, and lower-division courses received slightly higher average evaluations than upper-division courses. The relationship between age and evaluation scores was **very weak and negative (r = -0.048)**, indicating that age has almost no linear effect on evaluation scores. Gender and tenure also showed variation in evaluations, but no strong age-related pattern was observed overall.

Import libraries

```
import pandas as pd
import numpy as np
```

Load Data

```
df = pd.read_csv("/content/teachingratings_extracted.csv")
```

Question 1. Identify all duplicate cases using prof. Using all observations, find the average and standard deviation for age. Repeat the analysis by first filtering the data set to include one observation for each instructor with a total number of observations restricted to 94.

```
print("Number of observations:", len(df))
print("Number of unique instructors:", df['prof'].nunique())
```

```
Number of observations: 458
Number of unique instructors: 94
```

```
duplicate_cases = df[df.duplicated(subset='prof', keep=False)].sort_values('prof')
print("\nDuplicate cases:")
print(duplicate_cases[['prof', 'age', 'gender', 'eval', 'division', 'tenure']])
```

```
Duplicate cases:
     prof  age  gender  eval division tenure
3       1   36  female   4.4    upper    yes
1       1   36  female   3.7    upper    yes
0       1   36  female   4.3    upper    yes
2       1   36  female   3.6    upper    yes
5       2   59    male   4.0    upper    yes
..    ...  ...     ...   ...      ...    ...
445    93   32    male   3.2    lower    yes
450    94   42  female   4.1    lower    yes
447    94   42  female   3.3    upper    yes
448    94   42  female   3.3    upper    yes
449    94   42  female   3.2    upper    yes

[451 rows x 6 columns]
```

```
prof_counts = df['prof'].value_counts()
print("\nNumber of instructors with duplicate observations:",
      (prof_counts > 1).sum())
```

```
Number of instructors with duplicate observations: 87
```

```
print("\n--- All 458 observations ---")

mean_age = df['age'].mean()
sd_age = df['age'].std()
```

```
print("Mean age:", round(mean_age, 2))
print("Standard deviation:", round(sd_age, 2))
```

```
--- All 458 observations ---
Mean age: 48.32
Standard deviation: 9.82
```

```
df_94 = df.drop_duplicates(subset='prof', keep='first')
print("\n--- One observation per instructor ---")
print("Number of observations:", len(df_94))
mean_age_94 = df_94['age'].mean()
sd_age_94 = df_94['age'].std()

print("Mean age:", round(mean_age_94, 2))
print("Standard deviation:", round(sd_age_94, 2))
```

```
--- One observation per instructor ---
Number of observations: 94
Mean age: 47.55
Standard deviation: 10.26
```

Interpretation: When all 458 observations are used, the average instructor age is 48.32 years. After restricting the data to one observation per instructor, the average age becomes 47.55 years, with a slightly higher standard deviation of 10.26 years.

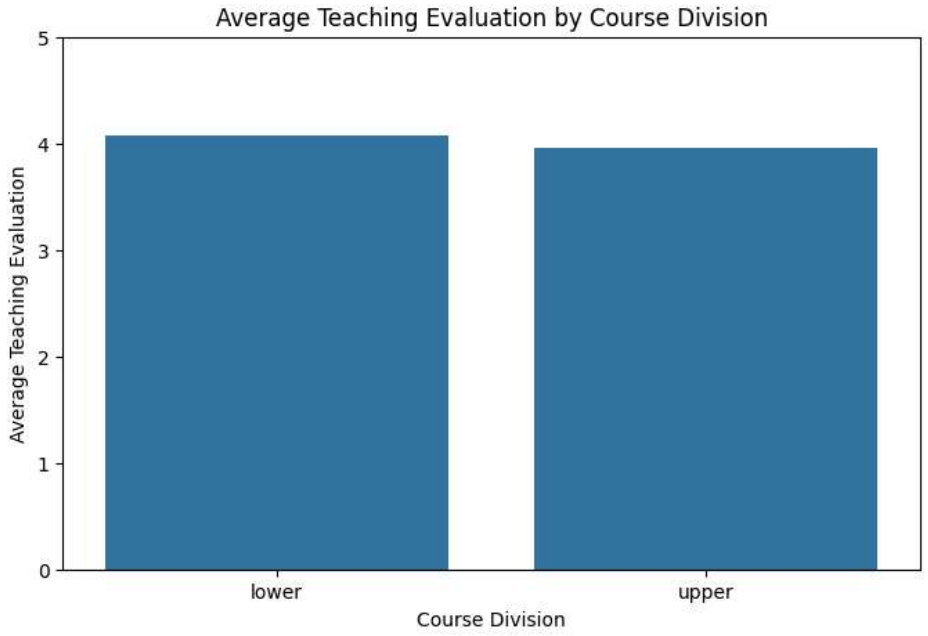
Question 2. Using a bar chart, demonstrate if instructors teaching lower-division courses receive higher average teaching evaluations.

```
import matplotlib.pyplot as plt
import seaborn as sns
```

```
division_avg = df.groupby('division')['eval'].mean().reset_index()
print(division_avg)
```

```
  division      eval
0    lower  4.080645
1    upper  3.955776
```

```
plt.figure(figsize=(8,5))
sns.barplot(data=division_avg, x='division', y='eval')
plt.title('Average Teaching Evaluation by Course Division')
plt.xlabel('Course Division')
plt.ylabel('Average Teaching Evaluation')
plt.ylim(0, 5)
plt.show()
```

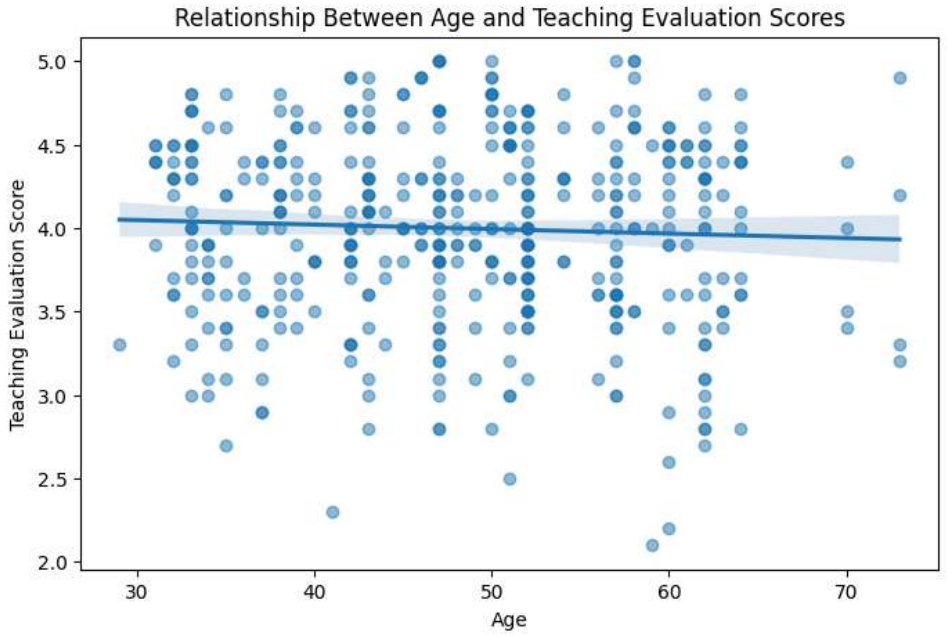


Interpretation

Lower-division instructors receive a slightly higher average teaching evaluation (4.08) compared with upper-division instructors (3.96).
Therefore, the bar chart provides evidence that evaluations are somewhat higher for lower-division courses.

Question 3. Plot the relationship between age and teaching evaluation scores.

```
plt.figure(figsize=(8,5))
sns.regplot(
    data=df,
    x='age',
    y='eval',
    scatter_kws={'alpha': 0.5},
    line_kws={}
)
plt.title('Relationship Between Age and Teaching Evaluation Scores')
plt.xlabel('Age')
plt.ylabel('Teaching Evaluation Score')
plt.show()
```




```
correlation = df['age'].corr(df['eval'])
print("Correlation between age and evaluation:", correlation)
```

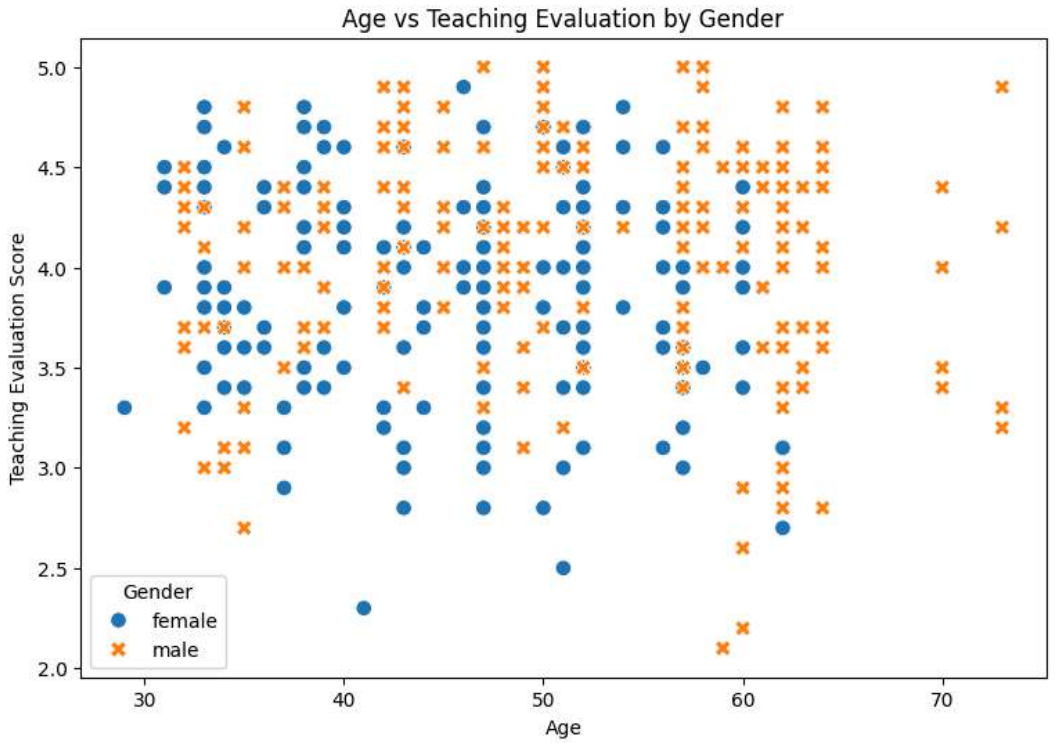
```
Correlation between age and evaluation: -0.047776570624846706
```

Interpretation

The regression plot shows a very weak negative relationship between age and teaching evaluation scores. The correlation coefficient (r = -0.048) is close to zero, indicating that instructor age has almost no linear relationship with teaching evaluation scores.

Question 4. Using gender-differentiated scatter plots, plot the relationship between age and teaching evaluation scores

```
plt.figure(figsize=(9,6))
sns.scatterplot(
    data=df,
    x='age',
    y='eval',
    hue='gender',
    style='gender',
    s=70
)
plt.title('Age vs Teaching Evaluation by Gender')
plt.xlabel('Age')
plt.ylabel('Teaching Evaluation Score')
plt.legend(title='Gender')
plt.show()
```

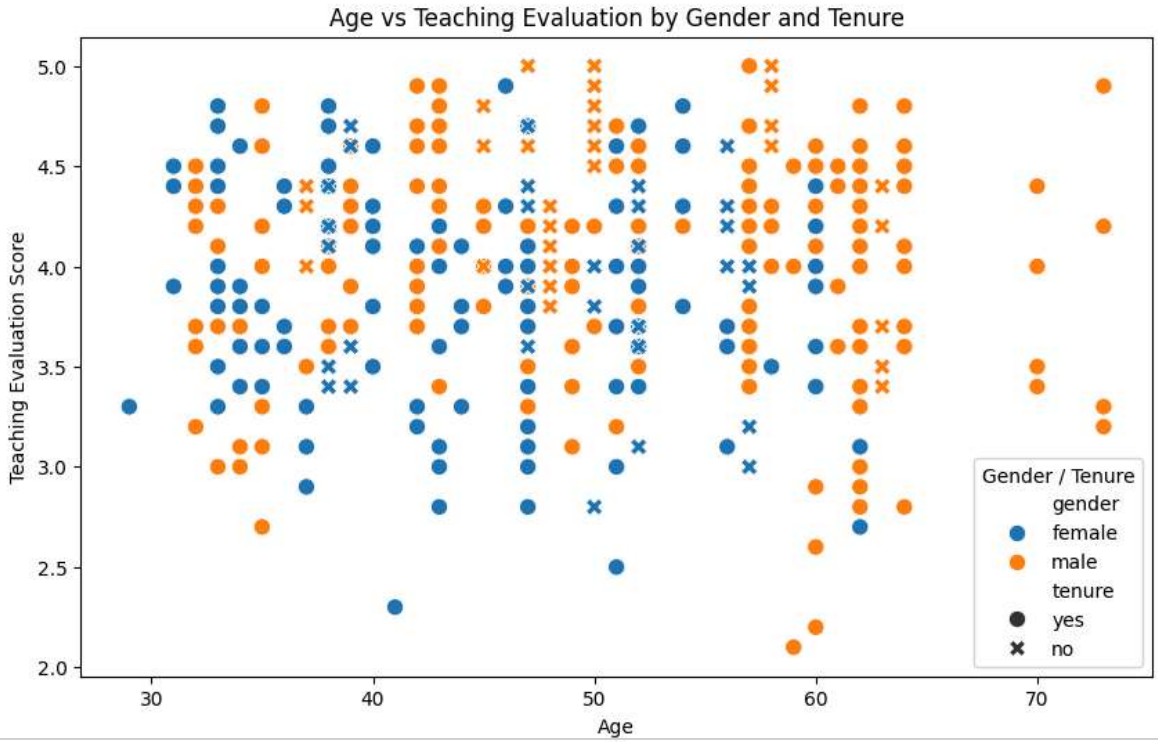


Interpretation

The gender-differentiated scatterplot allows us to compare the age-evaluation relationship for male and female instructors. Both genders show considerable variation in evaluation scores across different ages, and there does not appear to be a strong linear relationship between age and evaluation within either gender.

Question 5. Create a scatterplot of age and evaluation scores, differentiated by gender and tenure.

```
plt.figure(figsize=(10,6))
sns.scatterplot(
    data=df,
    x='age',
    y='eval',
    hue='gender',
    style='tenure',
    s=80
)
plt.title('Age vs Teaching Evaluation by Gender and Tenure')
plt.xlabel('Age')
plt.ylabel('Teaching Evaluation Score')
plt.legend(title='Gender / Tenure')
plt.show()
```



Interpretation

The scatterplot shows the relationship between instructor age and teaching evaluation while simultaneously distinguishing instructors by gender and tenure status. Evaluation scores vary considerably across ages and groups, indicating that age alone does not appear to strongly determine teaching evaluation scores. Gender and tenure provide additional group-level information that can be examined from the distribution of points.

```
Start coding or generate with AI.
```